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Day 8: IR OPSTACLE DETECTION

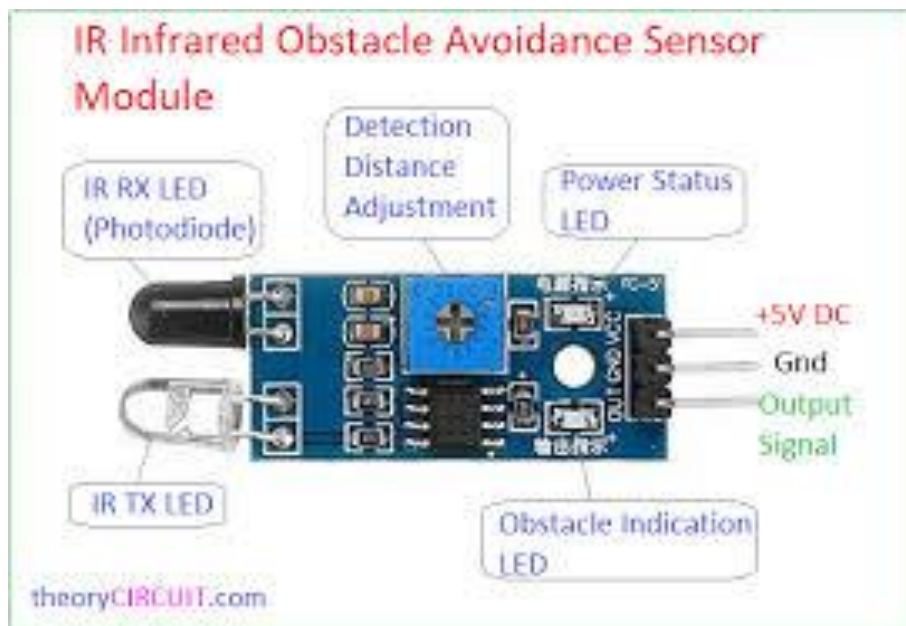
Date:28/07/2025

Domain: Embedded Systems and IoT

Topic: IR OPSTACLE DETECTION

What is an IR Obstacle Detection Sensor?

An **IR Obstacle Sensor** uses **infrared light** to detect objects in front of it. It consists of an IR LED (transmitter) and a photodiode (receiver). When an object is near, the IR light reflects back and is detected by the photodiode, which changes the sensor's output.



Why is it Used?

To detect nearby objects without contact, commonly used in

- Line-following robots
- Smart parking
- Hand sanitizers
- Obstacle-avoiding cars

Pin Configuration:

Pin Name	Function
----------	----------

VCC	Power Supply (5V)
GND	Ground
OUT	Digital Output Signal

How to Connect IR Sensor to Arduino Uno:

IR Sensor Pin	Arduino Uno Pin
VCC	5V
GND	GND
OUT	Digital Pin 7

Task 1: Detect Obstacle and Show via Serial Monitor

Objective:

Detect obstacle presence and show message on the Serial Monitor.

Code:

```
const int irSensor = 7;

void setup() {
  pinMode(irSensor, INPUT);
  Serial.begin(9600);
}

void loop() {
  int obstacle = digitalRead(irSensor);
  if (obstacle == LOW) {
    Serial.println("Obstacle Detected!");
  }
}
```

```

} else {

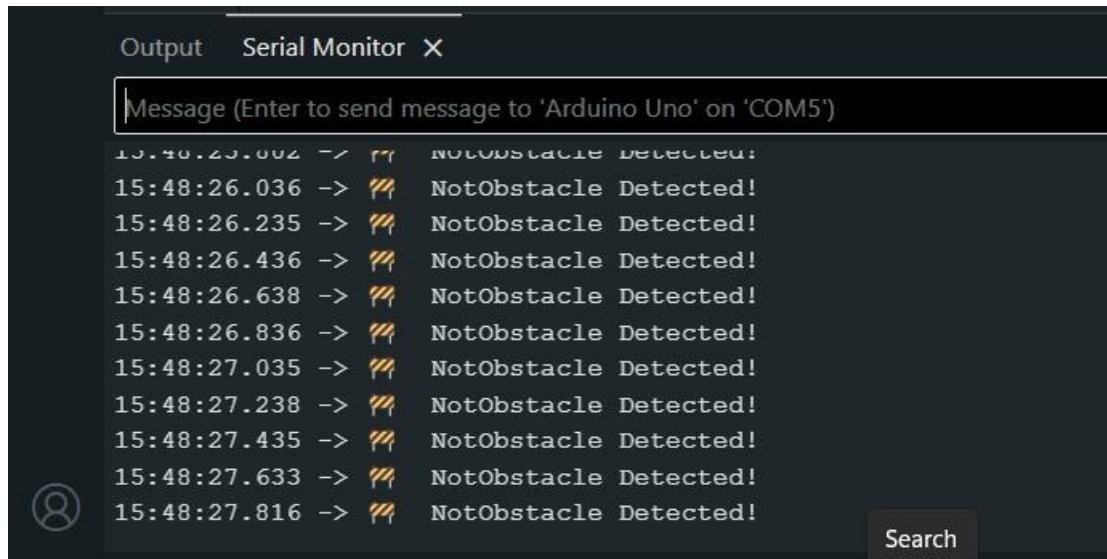
  Serial.println("No Obstacle");

}

delay(500);

}

```



Task 2: Turn ON LED if Obstacle Detected

Objective:

Light up the built-in LED (Pin 13) when an obstacle is detected.

Code:

```

const int irSensor = 7;

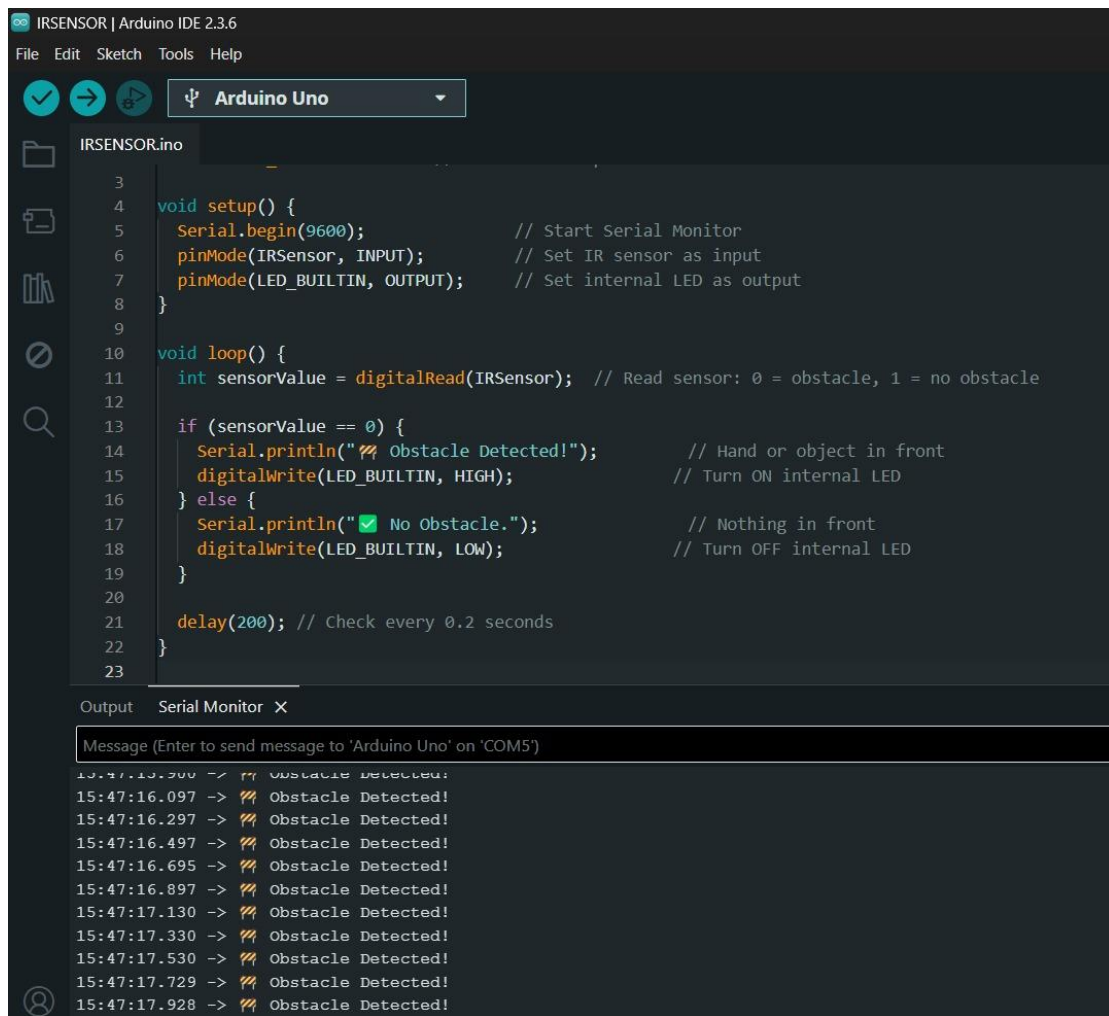
const int ledPin = 13;

void setup() {
  pinMode(irSensor, INPUT);
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int obstacle = digitalRead(irSensor);

```

```
if (obstacle == LOW) {  
  
    digitalWrite(ledPin, HIGH);  
  
    Serial.println("Obstacle Detected - LED ON");  
  
} else {  
  
    digitalWrite(ledPin, LOW);  
  
    Serial.println("No Obstacle - LED OFF");  
  
}  
  
delay(500);  
  
}
```



The screenshot displays the Arduino IDE 2.3.6 interface. The top menu bar includes File, Edit, Sketch, Tools, and Help. The toolbar shows icons for checking, running, and uploading code, along with a dropdown menu for the board set to 'Arduino Uno'. The left sidebar contains icons for file explorer, libraries, and search. The main editor window shows the 'IRSENSOR.ino' file with the following code:

```
3  
4 void setup() {  
5   Serial.begin(9600);           // Start Serial Monitor  
6   pinMode(IRSensor, INPUT);     // Set IR sensor as input  
7   pinMode(LED_BUILTIN, OUTPUT); // Set internal LED as output  
8 }  
9  
10 void loop() {  
11   int sensorValue = digitalRead(IRSensor); // Read sensor: 0 = obstacle, 1 = no obstacle  
12  
13   if (sensorValue == 0) {  
14     Serial.println("❌ Obstacle Detected!"); // Hand or object in front  
15     digitalWrite(LED_BUILTIN, HIGH);         // Turn ON internal LED  
16   } else {  
17     Serial.println("✅ No Obstacle.");        // Nothing in front  
18     digitalWrite(LED_BUILTIN, LOW);          // Turn OFF internal LED  
19   }  
20  
21   delay(200); // Check every 0.2 seconds  
22 }  
23
```

Below the editor is the 'Serial Monitor' tab, which shows a message input field and a list of received messages. The messages are all 'Obstacle Detected!' with timestamps ranging from 15:47:16.097 to 15:47:17.928.

Summary:

Feature	Details
Sensor	IR Osbtacle sensor
Microcontroller	Arduino Uno
Detection Method	Infrared Reflection
Output	HIGH(no obstacle),LOW (obstacle)
Task 1 Output	Serial Monitor
Task 2 Output	LED ON/OFF

Advantages:

- Low cost
- Quick response time
- Simple to interface with microcontrollers

Limitations:

- Affected by sunlight or reflective surfaces
- Short detection range (2-30 cm typical)

Applications:

- Obstacle detection in robots
- Automatic hand dryers
- Conveyor belt automation
- Smart bins